



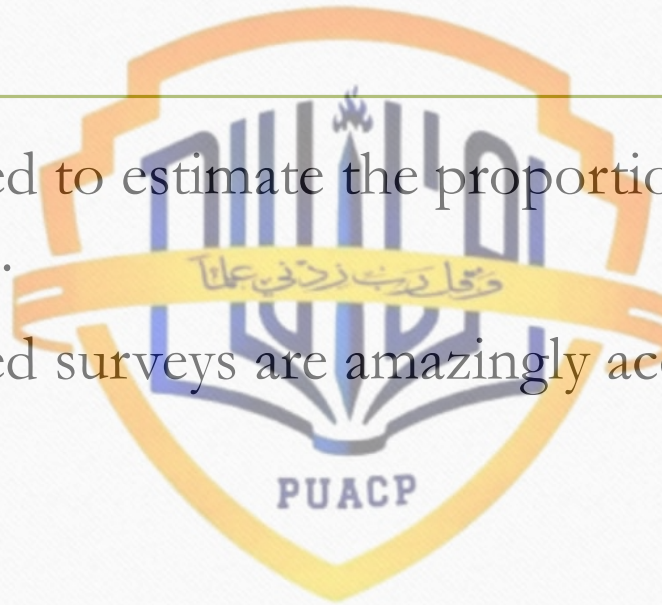
Lecture 5

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Sample Survey

- Sample surveys are used to estimate the proportion of people who have certain trait or opinion.
- The properly conducted surveys are amazingly accurate.



Sampling Error

- When a survey is used to find the proportion based on a sample of few thousand individuals, the obvious question is how close that proportion comes to the truth for entire population.
- This measure of accuracy in sample survey is margin of error/sampling errors
- Margin of error is the possible difference we allow between the result obtain from sample and true value of population.

Non Sampling Errors

- All the types of error other than sampling error when a survey is conducted are refer to non sampling errors

Such as

- Non Response
- Measurement Error
- Coverage Error
- Processing Error



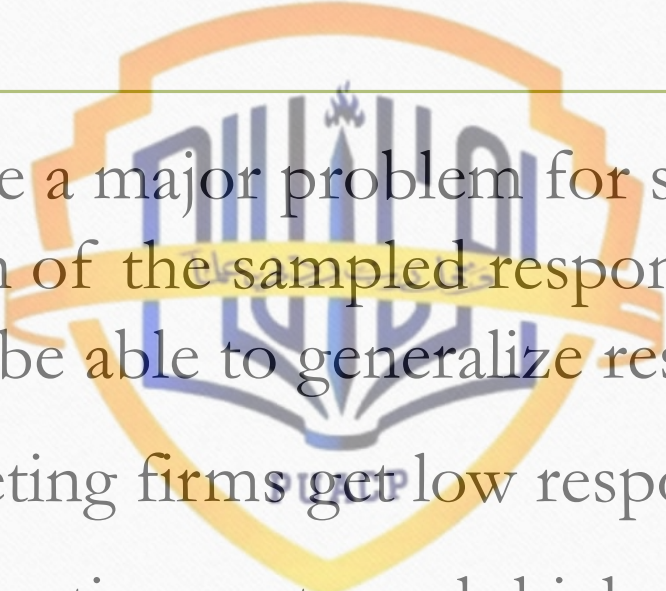
Non Response



Non-Response

- The proportion of people among those invited to participate in a survey who choose not to take part for any reasons.



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- Nonresponse can be a major problem for survey research because if a high proportion of the sampled respondents do not respond, researcher may not be able to generalize results.
 - Charities and marketing firms get low response rates.
 - Government organizations get much higher cooperation rates.

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- Nonresponse rates in survey vary greatly
 - For most academic organizations they range from twenty five to thirty three percent.
 - In commercial polls and campaign polls the nonresponse rates tends to be higher, reaching 50 percent.

Reasons

The reasons that respondents decline to participate could be that;

- They feel there are too many surveys
- A fear of crime and strangers
- A more hectic life style
- A loss of privacy
- A rising distrust of authority or government.
- The misuse of the survey to sell products
- Poor designed questionnaire
- Inadequate explanations of surveys to respondents



Respondents Vs Nonrespondents

- The people most likely to participate in surveys and polls are the most interested, informed and active in society.
- The people least likely to participate tend to be less informed about public events and less involved in socio-political activities.
 - They have a lower self-image, be more suspicious of strangers and distrust people of a different race.

Types of Nonresponse

- **Nonlocation** (could not find a sampled respondent)
- **Noncontact** (respondent was not at home or not reached after many attempts)
- **Ineligible** (respondent was reached but was not the proper age, race , sex, citizenship etc.)
- **Refusal to participate** (respondent was not willing to be interviewed)
- **Incomplete participation** (respondent stopped answering before the end or began answering every question with “do not know” or “no opinion”)

How to improve response?

To improve the overall survey response rate requires reducing each type of nonresponse.

- **Improving location** (it means better sampling frames, and maps or phone directories)
- **Improving contact** (it means to make many repeat calls, varying the time of day and lengthening the period to contact to try contacts)
- **Improving eligibility rates** (it can be done by careful respondent screening, better sample frame definition and multilingual interviewer)

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- **Decreasing refusals** (it can be achieved by sending letters in advance of an interview, offering, to reschedule interview, using small incentives or gifts, adjusting interviewer behavior by making eye contact, expressing sincerity, explaining the survey, emphasizing importance of the interview, clarifying promises of confidentiality.



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- **Increase Cooperation** (cooperation increases by showing a respondent that the survey topic will produce direct benefit to him or her)
 - Researcher increases cooperation among inner-city, low income racial-ethnic minorities by using journalistic style letter and by personal phone calls.

Measurement Error



Measurement Error

- This refers to the inaccuracies in recorded responses that occur when we incorrectly measure the characteristics of interest.



Sources of Measurement Error

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- **First:** Weakness in question wording.
 - A question should not be ambiguous.
 - A question should be clear and objectively presented in a natural manner.
 - Attention to detail is best precaution in most situation.

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- **Second:** Personal interviews and in telephone interviews
 - A halo effect in which the respondent feels obligation to please the interviewers
 - This type of error can be minimized by proper interviewer training.

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- Third: the effect on the part of respondent (response error)
 - These occurs when respondent do not understand the question properly.
 - This can be minimized by carefully scrutinizing the data and calling back respondent whose response seems unusual.
 - We can establish a program of random call backs for reliability of the responses.

Coverage Error



Coverage Error

- Coverage or selection error occurs when an inappropriate item is included in the survey.
- The key to proper sample selection is an adequate population frame of all the subjects from which the sample will be drawn.
- Coverage error results in a selection bias.
- If list is inadequate then certain group of subjects in the population were not properly included and any sample will not provide an estimate of the characteristics of actual population.

Processing Error



Processing Error

- These errors occur when data are incorrectly transferred from recording forms such as from questionnaire to computer files.



Thanks

